

R Code

```
1 ##### Practical 3 #####
2
3 # Title:
4 # Multivariate Normal Distribution and Conditional Distribution
5
6 # Objectives:
7 # 1. Generate multivariate normal data.
8 # 2. Perform conditional distribution calculations.
9 # 3. Apply the method to a real dataset.
10
11 #####
12 # 1. Load required package
13 #####
14
15 # Run the following line only once if MASS is not installed:
16 # install.packages("MASS")
17
18 library(MASS)
19
20 #####
21 # 2. Define mean vector and covariance matrix
22 #####
23
24 # Mean vector
25
26 mu <- c(10, 20, 30, 40)
27
28 mu
29
30 # Covariance matrix
31
32 Sigma <- matrix(
33   c(
34     9, 3, 2, 1,
35     3, 16, 4, 2,
36     2, 4, 25, 5,
37     1, 2, 5, 36
38   ),
39   nrow = 4,
40   byrow = TRUE
41 )
42
43 Sigma
44
45 #####
46 # 3. Generate multivariate normal data
47 #####
48
49 set.seed(123)
50
```

```
51  sim_data <- mvrnorm(  
52    n = 100000,  
53    mu = mu,  
54    Sigma = Sigma  
55  )  
56  
57  # Convert to data frame  
58  
59  sim_data <- as.data.frame(sim_data)  
60  
61  # Assign variable names  
62  
63  names(sim_data) <- c("X1", "X2", "X3", "X4")  
64  
65  # Display first few observations  
66  
67  head(sim_data)  
68  
69  #####  
70  # 4. Check sample mean and covariance matrix  
71  #####  
72  
73  # Sample mean vector  
74  
75  colMeans(sim_data)  
76  
77  # Sample covariance matrix  
78  
79  cov(sim_data)  
80  
81  #####  
82  # 5. Partition the mean vector  
83  #####  
84  
85  # Let:  
86  #  
87  # XA = X1  
88  #  
89  # XB = (X2, X3, X4)  
90  
91  mu_A <- mu[1]  
92  
93  mu_B <- mu[2:4]  
94  
95  mu_A  
96  mu_B  
97  
98  #####  
99  # 6. Partition the covariance matrix  
100 #####  
101  
102 # Sigma_AA  
103  
104 Sigma_AA <- matrix(  

```

```

105     Sigma[1, 1],
106     nrow = 1
107   )
108
109   # Sigma_AB
110
111   Sigma_AB <- matrix(
112     Sigma[1, 2:4],
113     nrow = 1
114   )
115
116   # Sigma_BA
117
118   Sigma_BA <- matrix(
119     Sigma[2:4, 1],
120     nrow = 3
121   )
122
123   # Sigma_BB
124
125   Sigma_BB <- Sigma[2:4, 2:4]
126
127   # Display the matrices
128
129   Sigma_AA
130   Sigma_AB
131   Sigma_BA
132   Sigma_BB
133
134   #####
135   # 7. Specify values of conditioning variables
136   #####
137
138   # X2 = 22
139   # X3 = 32
140   # X4 = 41
141
142   x_B <- c(22, 32, 41)
143
144   x_B
145
146   #####
147   # 8. Calculate conditional mean
148   #####
149
150   conditional_mean <- as.numeric(
151     mu_A +
152     Sigma_AB %*%
153     solve(Sigma_BB) %*%
154     (x_B - mu_B)
155   )
156
157   conditional_mean
158

```

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159 #####
160 # 9. Calculate conditional variance
161 #####
162
163 conditional_variance <- as.numeric(
164   Sigma_AA -
165   Sigma_AB %*%
166   solve(Sigma_BB) %*%
167   Sigma_BA
168 )
169
170 conditional_variance
171
172 #####
173 # 10. Calculate conditional standard deviation
174 #####
175
176 conditional_sd <- sqrt(conditional_variance)
177
178 conditional_sd
179
180 #####
181 # 11. Display the conditional distribution
182 #####
183
184 # X1 | (X2 = 22, X3 = 32, X4 = 41)
185 #
186 # follows a normal distribution with the above conditional mean
187 # and conditional variance.
188
189 cat(
190   "Conditional Mean =", conditional_mean, "\n",
191   "Conditional Variance =", conditional_variance, "\n",
192   "Conditional Standard Deviation =", conditional_sd, "\n"
193 )
194
195 #####
196 # 12. Generate observations from the conditional distribution
197 #####
198
199 # Generate 1000 observations from the conditional distribution
200
201 set.seed(123)
202
203 conditional_sample <- rnorm(
204   n = 1000,
205   mean = conditional_mean,
206   sd = conditional_sd
207 )
208
209 # Check sample mean
210
211 mean(conditional_sample)
212

```

```
213 # Check sample variance
214
215 var(conditional_sample)
216
217 #####
218 # 13. Compare theoretical and simulated results
219 #####
220
221 # Conditional mean
222
223 conditional_mean
224
225 mean(conditional_sample)
226
227 # Conditional variance
228
229 conditional_variance
230
231 var(conditional_sample)
232
233 #####
234 # 14. REAL DATA: USArrests
235 #####
236
237 # Load the built-in dataset
238
239 data("USArrests")
240
241 # Display first few observations
242
243 head(USArrests)
244
245 #####
246 # 15. Mean vector and covariance matrix
247 #####
248
249 # Mean vector
250
251 mu_us <- colMeans(USArrests)
252
253 mu_us
254
255 # Covariance matrix
256
257 Sigma_us <- cov(USArrests)
258
259 Sigma_us
260
261 #####
262 # 16. Conditional distribution of Murder
263 #   given Assault, UrbanPop and Rape
264 #####
265
266 # Let:
```

```

267 #
268 # XA = Murder
269 #
270 # XB = (Assault, UrbanPop, Rape)
271
272 #####
273 # Mean vector
274 #####
275
276 mu_A_us <- mu_us["Murder"]
277
278 mu_B_us <- mu_us[
279   c("Assault", "UrbanPop", "Rape")
280 ]
281
282 mu_A_us
283 mu_B_us
284
285 #####
286 # Covariance matrix blocks
287 #####
288
289 Sigma_AA_us <- matrix(
290   Sigma_us["Murder", "Murder"],
291   nrow = 1
292 )
293
294 Sigma_AB_us <- matrix(
295   Sigma_us[
296     "Murder",
297     c("Assault", "UrbanPop", "Rape")
298   ],
299   nrow = 1
300 )
301
302 Sigma_BA_us <- matrix(
303   Sigma_us[
304     c("Assault", "UrbanPop", "Rape"),
305     "Murder"
306   ],
307   nrow = 3
308 )
309
310 Sigma_BB_us <- Sigma_us[
311   c("Assault", "UrbanPop", "Rape"),
312   c("Assault", "UrbanPop", "Rape")
313 ]
314
315 #####
316 # 17. Specify conditioning values
317 #####
318
319 x_B_us <- c(
320   Assault = 200,

```

```

321     UrbanPop = 70,
322     Rape = 25
323 )
324
325 x_B_us
326
327 #####
328 # 18. Conditional mean of Murder
329 #####
330
331 cond_mean_us <- as.numeric(
332     mu_A_us +
333     Sigma_AB_us %*%
334     solve(Sigma_BB_us) %*%
335     (x_B_us - mu_B_us)
336 )
337
338 cond_mean_us
339
340 #####
341 # 19. Conditional variance of Murder
342 #####
343
344 cond_var_us <- as.numeric(
345     Sigma_AA_us -
346     Sigma_AB_us %*%
347     solve(Sigma_BB_us) %*%
348     Sigma_BA_us
349 )
350
351 cond_var_us
352
353 #####
354 # 20. Conditional standard deviation
355 #####
356
357 cond_sd_us <- sqrt(cond_var_us)
358
359 cond_sd_us
360
361 #####
362 # 21. Display the final conditional distribution
363 #####
364
365 cat(
366     "Conditional Mean of Murder =", cond_mean_us, "\n",
367     "Conditional Variance of Murder =", cond_var_us, "\n",
368     "Conditional Standard Deviation =", cond_sd_us, "\n"
369 )
370
371 #####
372 # PRACTICE Questions
373 #####
374

```

```
375 # Dataset: swiss
376
377 data("swiss")
378
379 # Display first few observations
380
381 head(swiss)
382
383 #####
384 # Q1. Mean vector and covariance matrix
385 #####
386
387 # Find the mean vector and covariance matrix.
388
389 colMeans(swiss)
390
391 cov(swiss)
392
393 #####
394 # Q2. Conditional distribution of Fertility
395 #####
396
397 # Consider:
398 #
399 # XA = Fertility
400 # XB = (Agriculture, Examination, Education)
401 #
402 # Find the conditional distribution of Fertility given
403 # Agriculture, Examination and Education.
404 #
405 # Use:
406 #
407 # Agriculture = 40
408 # Examination = 20
409 # Education   = 12
410 #
411 # Calculate:
412 # (a) Conditional mean
413 # (b) Conditional variance
414
415 #####
416 # Q3. Conditional distribution of Education
417 #####
418
419 # Consider:
420 #
421 # XA = Education
422 # XB = (Fertility, Examination, Catholic)
423 #
424 # Find the conditional mean and conditional variance of Education
425 # for suitable conditioning values.
426
427
428 # Q5. Your own example
```

```
429 #####
430
431 # Choose any one variable as XA and any three other variables as XB.
432 #
433 # Find the conditional mean and conditional variance for
434 # suitable conditioning values.
435
436
```